

**Question\_6:**

**Code:**

```
1 // There are n boys, m girls and k possible pairs. Find the maximum number of
2 // valid boy-girl pairs where each person can belong to at most one pair.
3 #include <iostream>
4 #include <cstring>
5 using namespace std;
6
7 #define MAXN 1005
8
9 int pairBoy[MAXN], pairGirl[MAXN];
10 int graph[MAXN][MAXN], degreeCount[MAXN], visited[MAXN];
11 void link(int u, int v)
12 {
13     graph[u][degreeCount[u]++] = v;
14 }
15 int dfs(int u)
16 {
17     for (int i = 0; i < degreeCount[u]; i++)
18     {
19         int v = graph[u][i];
20         if (!visited[v])
21         {
22             visited[v] = 1;
23             if (pairGirl[v] == -1 || dfs(pairGirl[v]))
24             {
25                 pairGirl[v] = u;
26                 pairBoy[u] = v;
27                 return 1;
28             }
29         }
30     }
31     return 0;
32 }
33 int main()
34 {
35     int n, m, k;
36     cin >> n >> m >> k;
37
38     for (int i = 0; i < MAXN; i++)
```

```
39     pairBoy[i] = pairGirl[i] = -1;
40
41     for (int i = 0; i < k; i++)
42     {
43         int u, v;
44         cin >> u >> v;
45         link(u, v);
46     }
47     int maxPairs = 0;
48     for (int i = 1; i <= n; i++)
49     {
50         memset(visited, 0, sizeof(visited));
51
52         if (dfs(i))
53             maxPairs++;
54     }
55     cout << maxPairs << endl;
56     for (int i = 1; i <= n; i++)
57         if (pairBoy[i] != -1)
58             cout << i << " " << pairBoy[i] << endl;
59     return 0;
60 }
```

**Output:**

**Test Case\_1:**

```
3 2 4
1 1
1 2
2 1
3 1
2
1 2
2 1
PS C:\Users\Prabin Yadav\
```

**Test Case\_2:**

```
3 3 4
1 1
1 2
2 3
3 2
3
1 1
2 3
3 2
PS C:\Users\Prabin Yadav\
```